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SliceInsightNet: Advanced Hybrid Image Segmentation for Spine Annotation in MRI

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***Abstract*—Segmentation in medical imaging is crucial for identifying and delineating anatomical structures, detecting pathological regions, and planning treatments. Deep learning techniques is the current state-of-the-art in several medical imaging segmentation applications. Three-dimensional (3D) data processing is known to efficient in handling changes in axial direction, however, it is computationally expensive. Processing images as a set of two-dimensional (2D) slices is more computationally efficient but with the cost of losing inter-slicing knowledge. This study focuses on the challenging task of spine segmentation, essential for diagnosing vertebral abnormalities and planning surgical interventions. We present SliceInsightNet, a novel hybrid approach that leverages both 2D and 3D segmentation techniques by utilizing sliced image data, optimizing computational efficiency and accuracy. Kullback–Leibler (KL) divergence is employed to calculate similarity metrics. The proposed model demonstrated slightly better results on unseen window configurations compared to UNet, and consistently outperformed UNet++ in both seen and unseen cases. Specifically, SliceInsightNet achieved a Dice Coefficient (DC) of 0.9712, Intersection over Union (IoU) of 0.9441, Pixel Accuracy of 0.9951, and F1 Boundary of 0.9848 for the same window configuration using the open access lumber spine segmentation dataset. For different window configurations, SliceInsightNet maintained competitive performance with a DC of 0.9171, IoU of 0.8475, Pixel Accuracy of 0.9881, and F1 Boundary of 0.9666. These results confirm SliceInsightNet's superior performance in spine segmentation, reinforcing its potential for broader clinical applications.**

***Index Terms*—Image Segmentation, Attention, Encoder, Slice, Kullback–Leibler (KL), UNet**

I. Introduction

Segmentation in medical imaging is a critical process that involves partitioning an image into meaningful regions. Medical segmentation is widely used in many applications, such as identifying and delineating anatomical structures, detecting pathological regions, and planning treatments. In particular, spine segmentation focuses on identifying the vertebrae and other spinal structures in imaging modalities such as MRI, CT, and X-rays. The spine's complex anatomy, with its intricate vertebral shapes and closely packed structures, presents significant challenges for segmentation. Spine segmentation aids in identifying vertebral abnormalities and planning surgical interventions.

Recent studies on medical segmentation have explored various models, including UNet, UNet++, UNetR, ViT Transformer, and ResUNet, to enhance performance and accuracy. UNet has been employed in studies such as those in [1], [2], demonstrating its use in heterogeneous datasets and attention mechanisms. UNet++, known for its nested architecture, has been used for improved feature extraction and multi- scale approaches [3]. Followup study improves UNet++ by taking advantage of full-scale skip connections and deep supervisions [4]. For transformer-based models, UNetR has demonstrated robust segmentation capabilities with enhanced attention modules [5]. ViT Transformer with hierarchical feature extraction and self-supervised pretraining with the Swin Transformer demonstrate a remarkable improvement in segmentation accuracy [6]. Furthermore, ResUNet represent an effective architecture for tasks like liver segmentation in CT images [7].

While two-dimensional (2D) segmentation provides valuable insights, three-dimensional (3D) segmentation offers a more comprehensive view by considering volumetric data, enabling more detailed visualization of anatomical structures. However, the main challenge with 3D segmentation is the high computational power required. To address both the issue of computational demand and the limitations in the accuracy of 2D segmentation, the proposed model, SliceInsightNet, works on 2D and 3D data simultaneously, combining the benefits of both approaches by extracting a window from the 3D data. This method teaches the model how each of the edge images (left and right) can help to segment the center images by combining them in an attention layer and then decoding the center image. Results are compared with the pre-trained models UNet and UNet++, demonstrating that the proposed model can compete with and excel beyond these models, particularly in handling unseen windows with different distributions.

This paper is structured as follows: we begin with the methodology section, detailing the data pre-processing steps and the experimental setup. Following this, we present the results and analysis of our experiments, comparing traditional training methods with K-fold cross-validation. Finally, we discuss the implications of our findings and conclude with future work and potential improvements.

II. Related Work

To tackle the challenges of spine segmentation, various techniques have been employed. For instance, Deep Convolutional

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Neural Networks (DCNNs), utilized for their ability to automatically extract and learn hierarchical features from medical images, have shown success in medical image segmentation. However, they often struggle with the anatomical complexities and deformities of the vertebrae [8]. The VerSe benchmark provides a comprehensive dataset and incorporating advanced techniques like iterative refinement and attention mechanisms and patch-based segmentation, which significantly improve segmentation accuracy and robustness. These innovations enable better handling of anatomical variations and enhance overall performance [9].

Other study has utilized UNet and its variants for improved segmentation accuracy [10]. For instance, Res50 UNet, an enhanced model combining feature pyramid network (FPN) and UNet with ResNet convolutional blocks and multi-scale feature extraction, has shown significant improvements in segmenting vertebral bodies in MRI scans. This method effectively handles rare anatomical variations and the lack of annotated datasets, significantly improving efficiency and accuracy in spine MRI segmentation. Furthermore, another study combines Atrous Spatial Pyramid Pooling (ASPP) with U-Net for spine MRI image segmentation, enhancing network feature extraction and achieving high accuracy in spine MRI image segmentation [11]. Additionally, SpineParseNet involves a 3D graph convolutional segmentation network (GCSN) for 3D coarse segmentation and a 2D residual U-Net (ResUNet) for 2D segmentation refinement, which improve the segmentation performance for volumetric MR images [12]. Later on, a method for 3D vertebral body segmentation in spine MRI images [13]. The process begins by correcting intensity to fix bias field artifacts and requires only minimal user assistance. Then, appearance-based maps of vertebral bodies guide the segmentation, which combines different techniques to achieve high accuracy and reliability across various imaging sequences and conditions.

Spine-GAN performs semantic segmentation of multiple spinal structures in MRIs using a recurrent generative adversarial network (GAN) [14]. Spine-GAN addresses the complexities of performing multiple tasks and targeting multiple spinal structures, achieving high segmentation accuracy. The model uses an atrous convolution autoencoder, a long short-term memory module, and a discriminative network to handle the variability and spatial correlations between normal and abnormal structures. An Axial-SpineGAN, which achieves simultaneous segmentation and diagnosis of multiple spinal structures on axial MRI images is introduced in [15] . This framework uses a combination of a generator, discriminator, and diagnostor to enhance segmentation and diagnostic accuracy.

An improved Attention U-Net model for lumbar spine segmentation, incorporating multilevel feature map fusion and introducing two residual modules to replace the original convolution blocks is presented in [16]. Additionally, the model employs a hybrid loss function during training, effectively enhancing accuracy and performance compared to existing algorithms. MRLN uses dilation convolution and LSTM modules to learn the relationships between vertebrae, enhancing segmentation accuracy [17]. It also incorporates a co-attention module to capture task correlations and employs an XOR loss function for efficient training.

An automatic method to segment 3D lumbar inter-vertebral discs from MRI scans using connected component analysis and statistical shape analysis is detailed in [18]. This method effectively addressing challenges such as partial volume effects, intensity in-homogeneity, and grey level overlap of different soft tissues in both normal and degenerated discs. DeepSPINE is an automated method for lumbar vertebral segmentation, disc-level designation, and spinal stenosis grading using deep learning, combining U-Net architecture for segmentation and a multi-task convolutional neural network for stenosis grading [19]. This method using natural language processing to extract ground-truth labels from radiology reports, improving the accuracy and efficiency of spinal stenosis diagnosis and enhancing radiology workflow.

Despite these advancements, there remain gaps in addressing the full complexity of spine segmentation, particularly in integrating 2D and 3D data efficiently. SliceInsideNet introduces a novel approach by combining 2D and 3D segmentation techniques, using the strengths of both methods to improve the accuracy and robustness of spine segmentation. This hybrid approach allows for better handling of complex anatomical structures and variations, providing a more comprehensive solution to the challenges in spine segmentation with low computational power.

III. Methodology

A. Data and Preprocessing

The dataset used in this study is a large, publicly available multi-center lumbar spine MRI dataset [20], which includes segmentations of vertebrae, inter-vertebral discs (IVDs), and the spinal canal. It consists of 447 sagittal T1w and T2w MRI series from 218 patients with a history of low back pain, collected from four different hospitals. The annotation process was iterative, starting with training a segmentation algorithm on a small subset of T2w images to semi-automatically segment the remaining images, which were then manually corrected and added to the training data. This dataset offers a diverse range of anatomical variations and pathological conditions, making it a valuable resource for training and evaluating segmentation models.

For preprocessing, the axes of some data were adjusted for consistency. To reduce import and processing time, all CPU processors were utilized, cutting the processing duration from 8.2 minutes to 1.2 minutes. Slices (windows) at positions 0.3, 0.5, and 0.7 were selected for analysis, and images were resized to 255×255 pixels for uniform dimensions. These specific slice positions were chosen to optimize the balance between capturing central and peripheral anatomical information. By selecting slices at these positions, the analysis ensures that both central and adjacent regions are well-represented, reducing the risk of missing critical variations while avoiding potential edge effects that might occur with slices too close

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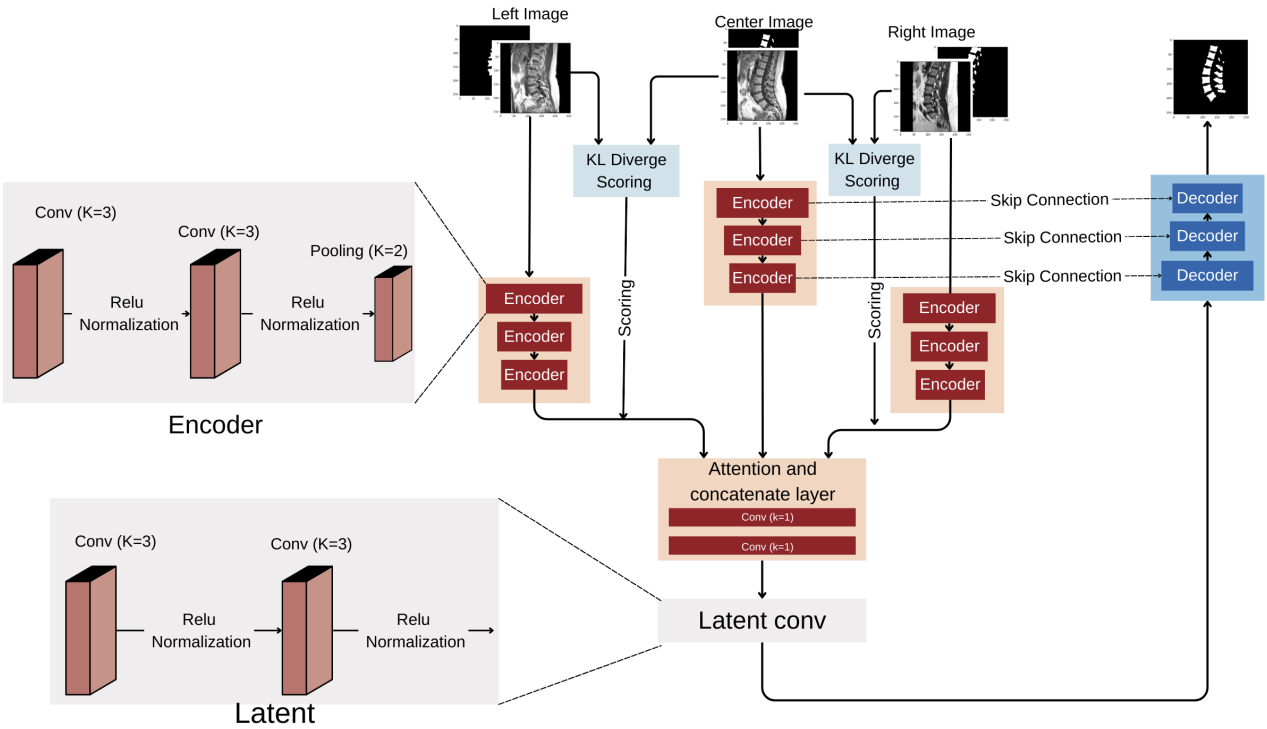

Fig. 1. Architecture of the proposed model SliceInsightNet.

to the outer boundaries. This strategy enhances the model's ability to accurately segment anatomical structures by providing a comprehensive view that integrates both detailed central features and contextual information from surrounding areas. For unseen data, slices at positions 0.4, 0.6, and 0.8 were used. This slight shift in slice selection for unseen data allows for testing the model's robustness and generalizability by introducing a controlled variation in the input data. A custom mask labeled vertebrae with indices from 1 to 6 as vertebrae (1) and other indices as non-vertebrae (0), resulting in binary classification. Finally, the pixel values of the images were normalized to a range between 0 and 1 by determining the maximum and minimum pixel values. For unseen data, slices at positions 0.4, 0.6, and 0.8 were used. While the slice sizes were kept consistent, the positions of the slices were shifted to test the model's robustness and generalizability by introducing a controlled variation in the input data.A custom mask labeled vertebrae with indices from 1 to 6 as vertebrae (1) and other indices as non-vertebrae (0), resulting in binary classification. Finally, the pixel values of the images were normalized to a range between 0 and 1 by determining the maximum and minimum pixel values.

B. Model Architecture

SliceInsightNet utilizes both 2D and 3D data to balance computational efficiency and effectiveness. The approach involves creating a windows by extracting three images from volumetric data slices: one from the left (index 0.3), one from the center (index 0.5), and one from the right (index 0.7). The objective is to examine how integrating information from the left and right images can enhance the segmentation accuracy of the center image.

Figure 1 illustrates the flow of the proposed model. First, each image (center and neighbors) and the corresponding mask are processed through three encoder steps. Each encoder consists of 2 convolutional layers with kernel size 3, 2 ReLU activation functions, 2 batch normalization layers, and a max pooling layer. Next, the outputs from each encoder block are fed into an attention and concatenation layer. The encoder outputs are combined by summing them, but the left and right images are given less importance compared to the center image through a scoring function. This function uses two convolutional layers with kernel size 1 to maintain spatial resolution, introduce non-linearity, and enhance computational efficiency and expressive power.

To calculate the score, we use the Kullback–Leibler (KL) divergence to measure the similarity between the previous (left), next (right), and center images. KL divergence quantifies the difference between the images' distributions, providing a normalized score between 0 and 1 to weight the contributions of each image. In our experiments, the ratios for left and right images are set to 1, 0.5 and 0.25, respectively. The attention output is then passed through a latent block, which includes 2 convolutional layers with kernel size 3, 2 ReLU functions,

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and 2 batch normalization layers. Finally, the decoder block receives two inputs: one from the latent convolutional block to reconstruct the segmented image and another from the skip connections at each hierarchical level to ensure comprehensive data transfer.

C. Evaluation Metrics

We use the Dice Coefficient to measure the overlap between the predicted and ground truth segmentation. It is defined as:

$$DC = \frac{2|A \cap B|}{|A| + |B|}, \tag{1}$$

where A and B represent the sets of predicted and ground truth positive pixels, respectively. This metric accounts for class imbalances, with a score of 1 indicating perfect overlap and a score of 0 indicating no overlap. Intersection over Union (IoU) measures the overlap between the predicted and ground truth segmentation as the ratio of the intersection to the union of the two sets:

$$IoU = \frac{|A \cap B|}{|A \cup B|}. \tag{2}$$

IoU provides a more stringent measure of accuracy, particularly useful in evaluating segmentation performance. Pixel accuracy is the ratio of correctly predicted pixels to the total number of pixels:

$$Acc = \frac{\text{umber of correctly predicted pixels}}{\text{Total number of pixels}}. \tag{3}$$

This metric provides a straightforward measure of overall segmentation performance. The F1 Boundary Score evaluates the accuracy of boundary predictions. It is the harmonic mean of precision and recall calculated at the boundaries of the segmented regions:

$$F_1 Scoere = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

This metric is particularly useful for assessing the accuracy of boundary delineation in segmentation tasks.

IV. Experiment and Result

We conducted two major tests to evaluate the effectiveness of SliceInsightNet. The first experiment used normal slice training (without cross-validation) methods with different ratios, while the second experiment employed K-fold cross-validation to ensure robustness and reliability of the results. Both experiments were conducted three times to validate the findings. Then, we compared SliceInsightNet with established architectures such as UNet and UNet++ by fine-tuning them with the data. For UNet and UNet++, we fine-tuned the models using the center image only, initially at index 0.5.

SliceInsightNet was designed to using the contextual information from neighboring slices to enhance segmentation accuracy, as previously mentioned. Additionally, we tested the model with unseen data to evaluate its performance in real-world scenarios. We conducted another test with an unseen window to assess the model's generalization capabilities. In this test, we altered the slice distribution, using a left image at index 0.4, a center image at index 0.6, and a right image at index 0.8. This approach was chosen to evaluate how well the models, including SliceInsightNet, could generalize when presented with a different slice distribution than what was used during training. This variation in slice positions tests the robustness of the model's ability to maintain segmentation accuracy despite changes in the input configuration. For the U-Net and U-Net++ models, we also tested the center image at index 0.6 to observe how shifts in the central slice position alone would affect their performance. By doing so, we could compare the sensitivity of these models to changes in the slice positions and highlight the adaptability of SliceInsightNet in maintaining consistent performance across varied configurations. Table I compares the model complexity, showing that SliceInsightNet has a significantly lower number of parameters compared to UNet and UNet++, which reduces computational costs while maintaining high segmentation accuracy.

TABLE I
Comparison of Model Architectures and Complexities

Component	UNet	UNet++	SliceInsightNet
Total Parameters	24,430,097	26,072,337	10,125,570
Encoder Structure	ResNet34 with 4 stages	ResNet34 with 4 stages + dense skip connections	Multiple encoders (3), each with 3 blocks
Attention Mechanism	No	No	Yes (SimpleAttention)
Decoder Structure	Standard UNet Decoder	UNet++ Decoder with dense skip connections	Simple decoder with 3 blocks

A. Slices without K-flod

By training using 50 epochs and applying early stopping for 10 steps, if the validation loss continues to increase, the training process is halted. This early stopping mechanism ensures that the model does not overfit and that the best performing model is selected based on validation performance. This training protocol was applied across all three experiments then average the result to ensure consistency and comparability. Results are shown in Table II.

B. K-fold cross-validation

Another study using cross-validation with data split into 5-folds, 10 epochs for each, and using the ratio 0.25. We then compared the outcomes with fine-tuned U-Net and U- Net++ models as depicted in Figure 2. The proposed model provided slightly better results on the unseen window configuration compared to U-Net and slightly better results in both seen and unseen cases compared to U-Net++. The average results (Table III) for U-Net++ were: Same Window for Dice, IOU, Pixel Accuracy, and F1 Boundary were 0.9642, 0.9318, 0.9941, and 0.9894, respectively; Different Window for Dice, IOU, Pixel Accuracy, and F1 Boundary were 0.9070, 0.8308, 0.9867, and 0.9522, respectively.

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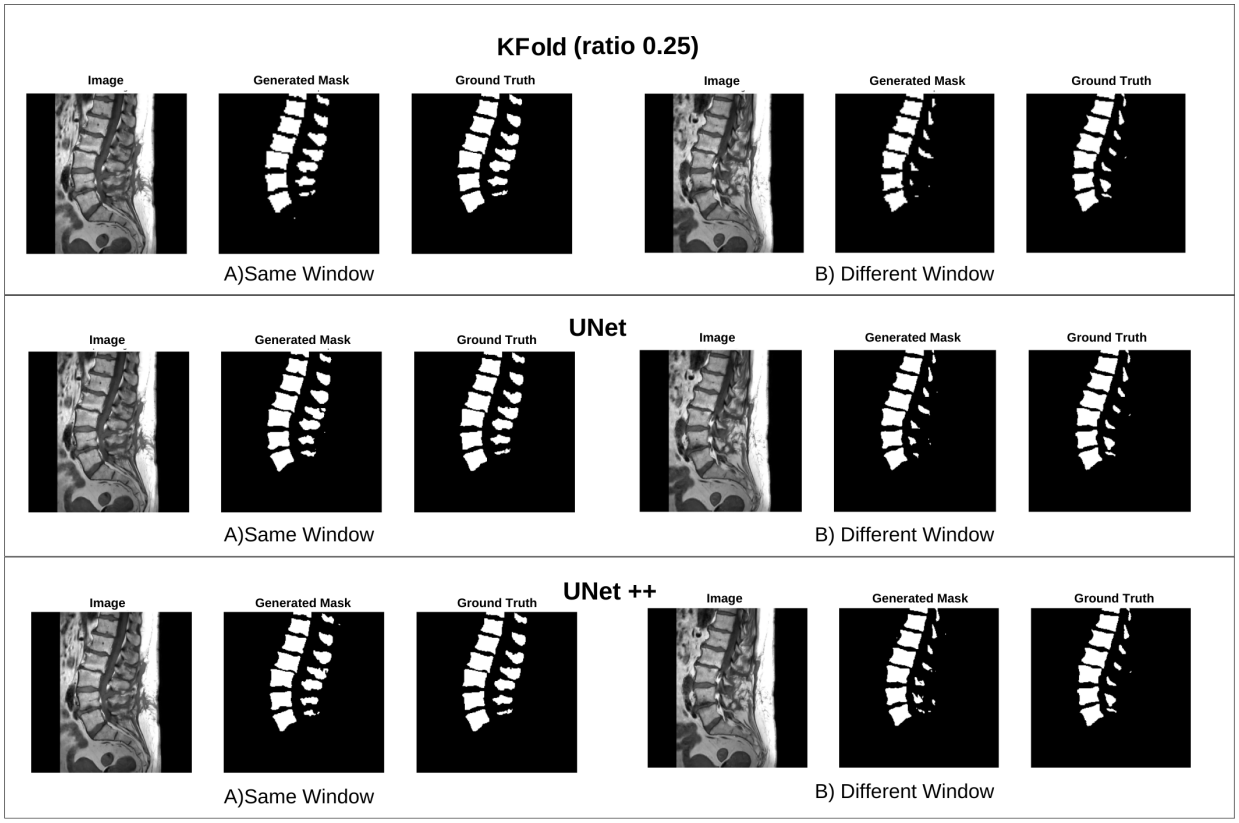

Fig. 2. Segmentation results comparing U-Net, U-Net++, and the proposed model using K-fold cross-validation.

TABLE II
RESULTS OF EACH EXPERIMENT ACROSS DIFFERENT RATIOS AND WINDOW. BOLD INDICATES SUPERIOR PERFORMANCE.

Exp.	Window	Measure	ratio=0.25	ratio=0.5	ratio=1.0
1	Same Slice Positions	DC	0.9408	**0.944**	0.9421
		IoU	0.8885	**0.8943**	0.8908
		Acc	0.9905	**0.9910**	0.9907
		F_1	0.9868	**0.9883**	0.9871
	Different Slice Positions	DC	0.9084	**0.9124**	0.9114
		IoU	0.8330	**0.8396**	0.8379
		Acc	0.9875	**0.988**	0.9879
		F_1	0.9619	0.9646	**0.9648**
2	Same Slice Positions	DC	**0.9425**	0.942	0.9392
		IoU	0.8915	**0.892**	0.8856
		Acc	**0.9902**	0.9892	0.9897
		F_1	**0.9868**	0.9858	0.9864
	Different Slice Positions	DC	**0.9049**	0.9001	0.9013
		IoU	**0.8267**	0.8231	0.8209
		Acc	**0.9862**	0.9831	0.9858
		F_1	**0.9593**	0.949	0.9562
3	Same Slice Positions	DC	**0.9384**	0.9354	0.9382
		IoU	**0.8841**	0.8789	0.8837
		Acc	**0.9893**	0.9887	**0.9893**
		F_1	0.9845	0.9816	**0.9847**
	Different Slice Positions	DC	**0.9002**	0.897	0.8953
		IoU	**0.8195**	0.814	0.8114
		Acc	**0.9856**	0.9851	0.985
		F_1	**0.9554**	0.9531	0.9509

SliceInsightNet's performance in segmenting unseen windows is notably more robust and accurate, as evidenced by higher DC, IoU, F1 score, and pixel accuracy scores across varied configurations. A key factor in this enhanced performance is the attention layer, which plays a crucial role by integrating features from the center image with dynamically adjusted contributions from the edge images. This mechanism mitigates the loss of inter-slicing knowledge by preserving essential contextual information from the surrounding slices. By selectively emphasizing the most relevant features, the attention layer adapts to variations in unseen windows, ensuring consistent segmentation quality. Results for the proposed model with K-fold cross-validation are shown in Table IV.

TABLE III
AVERAGE RESULTS FOR SLICEINSIGHTNET, UNET++ AND UNET WITH SAME AND DIFFERENT WINDOW

Window	Measure	SliceInsightNet	UNet++	UNet
Same Slice Positions	DC	0.9712	0.9642	**0.9735**
	IoU	0.9441	0.9318	**0.9484**
	Acc	0.9951	0.9941	0**0.9955**
	F_1	0.9848	0.9894	**0.9961**
Different Slice Positions	DC	**0.9171**	0.9070	0.9167
	IoU	**0.8475**	0.8308	0.8468
	Acc	0.**9881**	0.9867	0.9880
	F_1	**0.9666**	0.9522	0.9605

V. CONCLUSION

In this study, we introduced SliceInsightNet, a model designed for accurate central image segmentation using sliced image techniques. Using contextual information from neigh-

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TABLE IV
RESULTS OF EACH EXPERIMENT ACROSS K-FOLD (RATIO 0.25), UNET++, UNET

Exp.	Window	Measure	SliceInsightNet	UNet++	UNet
1	Same Slice Positions	DC	0.9659	0.9354	0.9773
		IoU	0.9340	0.8787	0.9557
		Acc	0.9944	0.9897	0.9963
		F_1	0.9980	0.9768	0.9977
	Different Slice Positions	DC	0.9191	0.8915	0.9232
		IoU	0.8511	0.8054	0.8578
		Acc	0.9889	0.9855	0.9893
		F_1	0.9666	0.9373	0.9658
2	Same Slice Positions	DC	0.9653	0.9748	0.9684
		IoU	0.9330	0.9510	0.9389
		Acc	0.9940	0.9957	0.9946
		F_1	0.9566	0.9946	0.9940
	Different Slice Positions	DC	0.9202	0.9133	0.913
		IoU	0.8524	0.8409	0.8403
		Acc	0.9883	0.9872	0.9872
		F_1	0.9694	0.9582	0.9566
3	Same Slice Positions	DC	0.9823	0.9824	0.9747
		IoU	0.9653	0.9656	0.9507
		Acc	0.9969	0.9969	0.9956
		F_1	0.9997	0.9969	0.9965
	Different Slice Positions	DC	0.9121	0.9163	0.914
		IoU	0.8390	0.8461	0.8423
		Acc	0.9872	0.9875	0.9874
		F_1	0.9639	0.9610	0.9590

boring slices, SliceInsightNet enhances segmentation accuracy, outperforming models like U-Net and U-Net++. Compared to fine-tuned U-Net and U-Net++ models, SliceInsightNet delivered better results, especially on unseen window configurations. This demonstrates its effectiveness in integrating contextual information from multiple slices for more accurate segmentation. The analysis indicates SliceInsightNet is highly suitable for scenarios involving unshown window data, which confirmed the model's robustness and generalization capabilities.

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